Cartier Divisors, Invertible Sheaves, and Holomorphic Line Bundles

Ji, Yong-hyeon

March 15, 2026

We cover the following topics in this note.

- Cartier Divisors
- Invertible Sheaves
- Holomorphic Line Bundles

Contents

1 Cartier Divisors $D = \{(U_\alpha, f_\alpha)\}$ **2**

2 Invertible Sheaf $\mathcal{O}_X(D)$ **4**

3 Cartier divisors and invertible sheaves: the precise dictionary **6**

1 Cartier Divisors $D = \{(U_\alpha, f_\alpha)\}$

Slogan. A Cartier divisor is a coherent way to package local winding-form data (i.e. local orders) so it is independent of choices of local equations.

Why Riemann surfaces and meromorphic functions? Let X be a Riemann surface. A holomorphic coordinate change $w = w(z)$ satisfies $w'(p) \neq 0$, so vanishing multiplicity is coordinate-invariant. Restricting to **meromorphic** functions excludes essential singularities (e.g. $e^{1/z}$), ensuring the order at a point is a **finite integer** (zeros and poles have finite order).

Order of a meromorphic function. Let $f \in \mathcal{M}_X(U)$ be nonzero and $p \in U$. Choose a local coordinate z with $z(p) = 0$. Then there exists a unique $k \in \mathbb{Z}$ and a holomorphic g on a neighborhood of p with $g(p) \neq 0$ such that

$$f(z) = z^k g(z).$$

Define

$$\mathrm{ord}_p(f) := k.$$

Equivalently (winding form / logarithmic derivative), for a small positively oriented loop γ around p,

$$\mathrm{ord}_p(f) = \frac{1}{2\pi i} \int_\gamma \frac{df}{f}.$$

Units do not change order. If $h \in \mathcal{O}_X^*(V)$ is holomorphic and nowhere vanishing near p, then $\mathrm{ord}_p(h) = 0$ and hence

$$\mathrm{ord}_p(hf) = \mathrm{ord}_p(h) + \mathrm{ord}_p(f) = \mathrm{ord}_p(f).$$

Analytically: on a sufficiently small simply connected neighborhood $W \ni p$, there exists a holomorphic logarithm H with $e^H = h$, so

$$\frac{dh}{h} = dH \quad \Rightarrow \quad \int_\gamma \frac{dh}{h} = 0$$

for every closed loop $\gamma \subset W$ ("FTC for holomorphic primitives").

Weil divisors (recording integer data at points). A (Weil) divisor on X is a finite formal sum

$$D = \sum_{p \in X} n_p \, p, \qquad n_p \in \mathbb{Z}, \;\; n_p = 0 \text{ for all but finitely many } p.$$

For a nonzero meromorphic function f, its **principal divisor** is

$$(f) := \sum_{p \in X} \mathrm{ord}_p(f)\, p.$$

Cartier divisors (local equations with overlap coherence). A Cartier divisor is given by an open cover $\{U_\alpha\}$ of X and meromorphic functions $f_\alpha \in \mathcal{M}_X^*(U_\alpha)$ such that on overlaps $U_{\alpha\beta} := U_\alpha \cap U_\beta$ one has

$$\frac{f_\alpha}{f_\beta} \in \mathcal{O}_X^*(U_{\alpha\beta}).$$

This condition **is exactly** what guarantees that the local orders glue: for any $p \in U_{\alpha\beta}$,

$$\mathrm{ord}_p(f_\alpha) = \mathrm{ord}_p\Big(\frac{f_\alpha}{f_\beta} \cdot f_\beta\Big) = \mathrm{ord}_p\Big(\frac{f_\alpha}{f_\beta}\Big) + \mathrm{ord}_p(f_\beta) = \mathrm{ord}_p(f_\beta),$$

since $\frac{f_\alpha}{f_\beta}$ is a unit.

Winding-form viewpoint on the overlap condition. On $U_{\alpha\beta}$,

$$\frac{df_\alpha}{f_\alpha} - \frac{df_\beta}{f_\beta} = \frac{d(f_\alpha/f_\beta)}{(f_\alpha/f_\beta)}.$$

If f_α/f_β is a holomorphic unit, then locally it has a holomorphic logarithm, so the right-hand side is exact; therefore its integral around a small loop is 0. Hence the winding integrals (and thus the orders) computed from df_α/f_α and df_β/f_β agree.

Conclusion. A Cartier divisor can be viewed as a collection of **local winding/ordering data** presented by meromorphic "local equations" $\{f_\alpha\}$, with the overlap rule $f_\alpha/f_\beta \in \mathcal{O}_X^*$ ensuring the induced integer ord_p is well-defined (choice-independent) at every point.

2 Invertible Sheaf $\mathcal{O}_X(D)$

Let X be a Riemann surface. A Cartier divisor D is given by an open cover $\{U_\alpha\}$ and $f_\alpha \in \mathcal{M}_X^*(U_\alpha)$ such that on overlaps $U_{\alpha\beta} := U_\alpha \cap U_\beta$ one has

$$\frac{f_\alpha}{f_\beta} \in \mathcal{O}_X^*(U_{\alpha\beta}).$$

Definition. Define a subsheaf $\mathcal{O}_X(D) \subset \mathcal{M}_X$ by declaring

$$\mathcal{O}_X(D)\big|_{U_\alpha} := f_\alpha^{-1}\,\mathcal{O}_X\big|_{U_\alpha}.$$

Equivalently, for $V \subset U_\alpha$,

$$\mathcal{O}_X(D)(V) = \{\, s \in \mathcal{M}_X(V) \;:\; f_\alpha s \in \mathcal{O}_X(V) \,\}.$$

Gluing on overlaps. On $U_{\alpha\beta}$,

$$f_\alpha^{-1}\mathcal{O}_X = f_\beta^{-1}\left(\frac{f_\beta}{f_\alpha}\right)\mathcal{O}_X = f_\beta^{-1}\mathcal{O}_X,$$

since $\frac{f_\beta}{f_\alpha} \in \mathcal{O}_X^*(U_{\alpha\beta})$. Hence $\mathcal{O}_X(D)$ is well-defined globally.

Invertibility. Multiplication by f_α yields an isomorphism

$$\mathcal{O}_X(D)\big|_{U_\alpha} \xrightarrow{\cdot f_\alpha} \mathcal{O}_X\big|_{U_\alpha},$$

so $\mathcal{O}_X(D)$ is locally free of rank 1, i.e. an invertible sheaf.

Transition functions. If $e_\alpha := f_\alpha^{-1}$ is a local generator of $\mathcal{O}_X(D)$ on U_α, then on $U_{\alpha\beta}$,

$$e_\alpha = \left(\frac{f_\beta}{f_\alpha}\right) e_\beta,$$

so the transition functions are $g_{\alpha\beta} := \frac{f_\beta}{f_\alpha} \in \mathcal{O}_X^*(U_{\alpha\beta})$.

Logarithmic differentials (winding-form view). On $U_{\alpha\beta}$,

$$\frac{df_\alpha}{f_\alpha} - \frac{df_\beta}{f_\beta} = \frac{d(f_\alpha/f_\beta)}{(f_\alpha/f_\beta)}.$$

Since $f_\alpha / f_\beta \in \mathcal{O}_X^*$, locally it admits a holomorphic logarithm, so the right-hand side is exact and its integral over small loops is 0.

3 Cartier divisors and invertible sheaves: the precise dictionary

Let X be a Riemann surface. Write $\mathcal{O}_X$ for the sheaf of holomorphic functions, $\mathcal{O}_X^*$ for nowhere-vanishing holomorphic functions (units), $\mathcal{M}_X$ for meromorphic functions, and $\mathcal{M}_X^*$ for nonzero meromorphic functions.

1) Cartier divisors as a quotient sheaf

A **Cartier divisor** can be defined as a global section of the quotient sheaf

$$\mathcal{M}_X^*/\mathcal{O}_X^*.$$

Concretely, it is given by a cover $\{U_\alpha\}$ and $f_\alpha \in \mathcal{M}_X^*(U_\alpha)$ such that

$$\frac{f_\alpha}{f_\beta} \in \mathcal{O}_X^*(U_{\alpha\beta}) \quad \text{on } U_{\alpha\beta} := U_\alpha \cap U_\beta,$$

and two such collections $\{f_\alpha\}$ and $\{f'_\alpha\}$ define the same Cartier divisor iff $f'_\alpha/f_\alpha \in \mathcal{O}_X^*(U_\alpha)$ for all α.

2) From a Cartier divisor D to the invertible sheaf $\mathcal{O}_X(D)$

Given Cartier data $\{(U_\alpha, f_\alpha)\}$, define

$$\mathcal{O}_X(D)\big|_{U_\alpha} := f_\alpha^{-1}\,\mathcal{O}_X\big|_{U_\alpha} \subset \mathcal{M}_X\big|_{U_\alpha}.$$

Equivalently, for $V \subset U_\alpha$,

$$\mathcal{O}_X(D)(V) = \{\, s \in \mathcal{M}_X(V) \;:\; f_\alpha s \in \mathcal{O}_X(V) \,\}.$$

On overlaps,

$$f_\alpha^{-1}\mathcal{O}_X = f_\beta^{-1}\left(\frac{f_\beta}{f_\alpha}\right)\mathcal{O}_X = f_\beta^{-1}\mathcal{O}_X \quad \text{since } \frac{f_\beta}{f_\alpha} \in \mathcal{O}_X^*,$$

so the local definitions glue. Moreover multiplication by f_α gives an isomorphism

$$\mathcal{O}_X(D)\big|_{U_\alpha} \xrightarrow{\cdot f_\alpha} \mathcal{O}_X\big|_{U_\alpha},$$

hence $\mathcal{O}_X(D)$ is locally free of rank 1, i.e. an invertible sheaf.

3) From an invertible sheaf to a Cartier divisor

Let $\mathcal{L}$ be an invertible sheaf on X. Choose a cover $\{U_\alpha\}$ and local frames e_α identifying

$$\mathcal{L}|_{U_\alpha} \cong \mathcal{O}_X|_{U_\alpha}.$$

On overlaps $U_{\alpha\beta}$ there are transition functions

$$e_\alpha = g_{\alpha\beta}\, e_\beta, \qquad g_{\alpha\beta} \in \mathcal{O}_X^*(U_{\alpha\beta}),$$

and $\{g_{\alpha\beta}\}$ is a Čech 1-cocycle with values in $\mathcal{O}_X^*$. Now pick meromorphic trivializations s_α of $\mathcal{L}$ on U_α (nonzero meromorphic sections); write $s_\alpha = f_\alpha e_\alpha$ with $f_\alpha \in \mathcal{M}_X^*(U_\alpha)$. Then on overlaps,

$$\frac{f_\alpha}{f_\beta} = g_{\alpha\beta} \in \mathcal{O}_X^*(U_{\alpha\beta}),$$

so $\{f_\alpha\}$ defines a Cartier divisor. (Intuitively: $\mathcal{L}$ becomes meromorphically trivial, and the failure of holomorphic triviality is measured by the divisor.)

4) Principal divisors correspond to trivial line bundles

If $D = (f)$ is principal (coming from a global $f \in \mathcal{M}_X^*(X)$), take $f_\alpha = f|_{U_\alpha}$. Then

$$\mathcal{O}_X((f)) = f^{-1}\mathcal{O}_X \cong \mathcal{O}_X$$

via multiplication by f. Thus principal divisors map to the trivial bundle.

5) Picard group and the exact-sequence explanation

The **Picard group** is the group of isomorphism classes of invertible sheaves:

$$\mathrm{Pic}(X) := \{\text{invertible sheaves on } X\}/\cong,$$

with tensor product as the group law.

There is a short exact sequence of sheaves of abelian groups

$$1 \longrightarrow \mathcal{O}_X^* \longrightarrow \mathcal{M}_X^* \longrightarrow \mathcal{M}_X^*/\mathcal{O}_X^* \longrightarrow 1.$$

Taking cohomology yields a connecting homomorphism

$$\delta : H^0\left(X, \mathcal{M}_X^*/\mathcal{O}_X^*\right) \longrightarrow H^1(X, \mathcal{O}_X^*).$$

Here

$$H^0\left(X, \mathcal{M}_X^*/\mathcal{O}_X^*\right) \cong \{\text{Cartier divisors on } X\}, \qquad H^1(X, \mathcal{O}_X^*) \cong \mathrm{Pic}(X).$$

Under these identifications, $\delta(D)$ is exactly the isomorphism class of the invertible sheaf $\mathcal{O}_X(D)$.

Moreover, $\ker(\delta)$ consists of divisors that come from global meromorphic functions (principal

divisors), so one gets the fundamental identification

$$\mathrm{CaCl}(X) := \frac{\{\text{Cartier divisors}\}}{\{\text{principal Cartier divisors}\}} \cong \mathrm{Pic}(X), \qquad [D] \longmapsto [\mathcal{O}_X(D)].$$

6) How this matches your winding-form intuition

On overlaps $U_{\alpha\beta}$, put $u_{\alpha\beta} := f_\alpha / f_\beta \in \mathcal{O}_X^*$. Then

$$\frac{df_\alpha}{f_\alpha} - \frac{df_\beta}{f_\beta} = \frac{du_{\alpha\beta}}{u_{\alpha\beta}}.$$

Locally (on a small simply connected set) $u_{\alpha\beta} = e^{h_{\alpha\beta}}$ for a holomorphic $h_{\alpha\beta}$, so

$$\frac{du_{\alpha\beta}}{u_{\alpha\beta}} = dh_{\alpha\beta}$$

is exact. Hence small-loop integrals of $\frac{df_\alpha}{f_\alpha}$ agree across overlaps, so the local winding/order data glue. The same units $u_{\alpha\beta}$ are also exactly the transition functions of the line bundle $\mathcal{O}_X(D)$:

$$e_\alpha = u_{\beta\alpha}\, e_\beta \quad \text{with} \quad u_{\beta\alpha} = f_\beta / f_\alpha \in \mathcal{O}_X^*.$$

Sheaves and the definition of $\mathcal{O}_X(D)$

Definition (presheaf and sheaf). Let X be a topological space. A **presheaf** $\mathcal{F}$ (of sets / abelian groups / rings / modules) assigns to each open set $U \subset X$ an object $\mathcal{F}(U)$ and to each inclusion $V \subset U$ a restriction map $\rho_{UV} : \mathcal{F}(U) \to \mathcal{F}(V)$, $s \mapsto s|_V$, such that

$$s|_U = s, \qquad (s|_V)|_W = s|_W \quad (W \subset V \subset U).$$

A presheaf is a **sheaf** if for every open cover $U = \bigcup_i U_i$ the following hold:

(S1) Locality. If $s, t \in \mathcal{F}(U)$ and $s|_{U_i} = t|_{U_i}$ for all i, then $s = t$.
(S2) Gluing. If $s_i \in \mathcal{F}(U_i)$ satisfy $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j, then there exists a unique $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i.

Setup. Let X be a Riemann surface. Let $\mathcal{O}_X$ be the sheaf of holomorphic functions and $\mathcal{M}_X$ the sheaf of meromorphic functions. A Cartier divisor D is given by an open cover $\{U_\alpha\}$ and $f_\alpha \in \mathcal{M}_X^*(U_\alpha)$ such that on overlaps $U_{\alpha\beta} := U_\alpha \cap U_\beta$,

$$\frac{f_\alpha}{f_\beta} \in \mathcal{O}_X^*(U_{\alpha\beta}).$$

Definition (the sheaf $\mathcal{O}_X(D)$). Define a presheaf $\mathcal{O}_X(D) \subset \mathcal{M}_X$ by

$$\mathcal{O}_X(D)(V) := \left\{ s \in \mathcal{M}_X(V) \ : \ (f_\alpha s)\big|_{V \cap U_\alpha} \in \mathcal{O}_X(V \cap U_\alpha) \ \forall \alpha \right\}$$

for every open $V \subset X$. Restriction maps are the restrictions in $\mathcal{M}_X$: $s \mapsto s|_W$ for $W \subset V$.

Lemma (well-defined on overlaps). On $V \cap U_{\alpha\beta}$, the condition $(f_\alpha s) \in \mathcal{O}_X$ is equivalent to $(f_\beta s) \in \mathcal{O}_X$.

Proof. On $V \cap U_{\alpha\beta}$,

$$f_\alpha s = \left(\frac{f_\alpha}{f_\beta}\right)(f_\beta s).$$

Since $\frac{f_\alpha}{f_\beta} \in \mathcal{O}_X^*(U_{\alpha\beta})$, multiplication by this factor preserves holomorphicity, hence $f_\alpha s$ is holomorphic iff $f_\beta s$ is holomorphic. □

Proposition (sheaf property). The presheaf $\mathcal{O}_X(D)$ is a sheaf (indeed, an $\mathcal{O}_X$-module subsheaf of $\mathcal{M}_X$).

Proof. Let $V = \bigcup_i V_i$ be an open cover.

Locality (S1). If $s, t \in \mathcal{O}_X(D)(V)$ and $s|_{V_i} = t|_{V_i}$ for all i, then $s = t$ in $\mathcal{M}_X(V)$ because $\mathcal{M}_X$ is a sheaf. Hence $s = t$ in $\mathcal{O}_X(D)(V)$.

Gluing (S2). Let $s_i \in \mathcal{O}_X(D)(V_i)$ satisfy $s_i = s_j$ on $V_i \cap V_j$. Since $\mathcal{M}_X$ is a sheaf, there exists a unique $s \in \mathcal{M}_X(V)$ with $s|_{V_i} = s_i$. We claim $s \in \mathcal{O}_X(D)(V)$.

Fix α. On each $V_i \cap U_\alpha$,

$$(f_\alpha s)|_{V_i \cap U_\alpha} = f_\alpha (s|_{V_i \cap U_\alpha}) = f_\alpha (s_i|_{V_i \cap U_\alpha}),$$

and the right-hand side lies in $\mathcal{O}_X(V_i \cap U_\alpha)$ because $s_i \in \mathcal{O}_X(D)(V_i)$. Moreover these holomorphic functions agree on overlaps because the s_i agree. Since $\mathcal{O}_X$ is a sheaf, they glue to a holomorphic function on $V \cap U_\alpha$. Thus $(f_\alpha s)|_{V \cap U_\alpha} \in \mathcal{O}_X(V \cap U_\alpha)$ for all α, i.e. $s \in \mathcal{O}_X(D)(V)$.

Uniqueness follows from uniqueness in $\mathcal{M}_X$. Therefore $\mathcal{O}_X(D)$ satisfies (S1) and (S2), hence is a sheaf. □

Sheaves and $\mathcal{O}_{\mathbf{CP}^1}(D)$ on $\mathbf{CP}^1$

Definition (sheaf). Let X be a topological space. A **presheaf** $\mathcal{F}$ (of sets / abelian groups / rings / modules) assigns to every open set $U \subset X$ an object $\mathcal{F}(U)$ and to every inclusion $V \subset U$ a restriction map

$$\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V), \qquad s \mapsto s|_V,$$

such that $\mathrm{res}_{U,U} = \mathrm{id}$ and $\mathrm{res}_{V,W} \circ \mathrm{res}_{U,V} = \mathrm{res}_{U,W}$ for $W \subset V \subset U$. A presheaf is a **sheaf** if for every open cover $U = \bigcup_i U_i$:

(S1) Locality. If $s, t \in \mathcal{F}(U)$ and $s|_{U_i} = t|_{U_i}$ for all i, then $s = t$.
(S2) Gluing. If $s_i \in \mathcal{F}(U_i)$ satisfy $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j, then there exists a unique $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i.

The standard cover of $\mathbf{CP}^1$. Let $X = \mathbf{CP}^1$ with the standard affine cover

$$U_0 = \{[z_0 : z_1] \in \mathbf{CP}^1 : \ z_0 \neq 0\}, \qquad U_\infty = \{[z_0 : z_1] \in \mathbf{CP}^1 : \ z_1 \neq 0\}.$$

On U_0 use the coordinate $z = z_1/z_0 \in \mathbf{C}$, and on U_∞ use $w = z_0/z_1 \in \mathbf{C}$. On the overlap $U_0 \cap U_\infty$ one has $w = 1/z$ and $U_0 \cap U_\infty \simeq \mathbf{C}^*$.

Setup (Cartier divisor). Let D be a Cartier divisor on X given by meromorphic functions

$$f_0 \in \mathcal{M}_X^*(U_0), \qquad f_\infty \in \mathcal{M}_X^*(U_\infty),$$

such that on the overlap

$$\frac{f_0}{f_\infty} \in \mathcal{O}_X^*(U_0 \cap U_\infty).$$

Definition (the presheaf $\mathcal{O}_X(D)$). For each open set $V \subset X$, define

$$\mathcal{O}_X(D)(V) := \left\{ s \in \mathcal{M}_X(V) : (f_0 s)\big|_{V \cap U_0} \in \mathcal{O}_X(V \cap U_0),\ (f_\infty s)\big|_{V \cap U_\infty} \in \mathcal{O}_X(V \cap U_\infty) \right\}.$$

Restriction maps are the restrictions in $\mathcal{M}_X$: for $W \subset V$, $\mathrm{res}_{V,W}(s) = s|_W$.

Lemma (overlap consistency). On $V \cap U_0 \cap U_\infty$, the condition $(f_0 s) \in \mathcal{O}_X$ is equivalent to $(f_\infty s) \in \mathcal{O}_X$.

Proof. On $U_0 \cap U_\infty$ we have $f_0 = u\, f_\infty$ where $u := f_0/f_\infty \in \mathcal{O}_X^*(U_0 \cap U_\infty)$. Hence

$$f_0 s = u\, (f_\infty s).$$

Since u is holomorphic and nowhere vanishing, multiplication by u preserves holomorphicity. Therefore $f_0 s$ is holomorphic iff $f_\infty s$ is holomorphic. □

Proposition (sheaf property on $\mathbf{CP}^1$). The presheaf $\mathcal{O}_X(D)$ is a sheaf (indeed, an $\mathcal{O}_X$-module subsheaf of $\mathcal{M}_X$).

Proof. Let $V = \bigcup_i V_i$ be an open cover.

Locality (S1). If $s, t \in \mathcal{O}_X(D)(V)$ and $s|_{V_i} = t|_{V_i}$ for all i, then $s = t$ in $\mathcal{M}_X(V)$ because $\mathcal{M}_X$ is a sheaf. Hence $s = t$ in $\mathcal{O}_X(D)(V)$.

Gluing (S2). Let $s_i \in \mathcal{O}_X(D)(V_i)$ satisfy $s_i = s_j$ on $V_i \cap V_j$. Since $\mathcal{M}_X$ is a sheaf, there exists a unique $s \in \mathcal{M}_X(V)$ with $s|_{V_i} = s_i$. We show $s \in \mathcal{O}_X(D)(V)$.

On $V \cap U_0$, the open cover $\{V_i \cap U_0\}_i$ satisfies

$$(f_0 s)|_{V_i \cap U_0} = f_0(s|_{V_i \cap U_0}) = f_0(s_i|_{V_i \cap U_0}) \in \mathcal{O}_X(V_i \cap U_0),$$

because $s_i \in \mathcal{O}_X(D)(V_i)$. These holomorphic functions agree on overlaps (since the s_i agree), hence glue (by the sheaf property of $\mathcal{O}_X$) to a holomorphic function on $V \cap U_0$. Thus $(f_0 s)|_{V \cap U_0} \in \mathcal{O}_X(V \cap U_0)$. The same argument on $V \cap U_\infty$ shows $(f_\infty s)|_{V \cap U_\infty} \in \mathcal{O}_X(V \cap U_\infty)$. Therefore $s \in \mathcal{O}_X(D)(V)$.

Uniqueness is inherited from uniqueness in $\mathcal{M}_X$. □

Example: $D = n[\infty]$ gives $\mathcal{O}(n)$. Let $D = n[\infty]$. Take $f_0 = 1$ on U_0 and $f_\infty = w^n$ on U_∞. On $U_0 \cap U_\infty$,

$$\frac{f_0}{f_\infty} = \frac{1}{w^n} = z^n \in \mathcal{O}_X^*(\mathbf{C}^*),$$

so this is Cartier data. Then $s \in \mathcal{O}(n)(X)$ iff s is holomorphic on $U_0 \simeq \mathbf{C}$ and $w^n s$ is holomorphic at $w = 0$ (i.e. at ∞), hence s is a polynomial in z of degree $\leq n$. Consequently,

$$H^0(\mathbf{CP}^1, \mathcal{O}(n)) \cong \langle 1, z, \ldots, z^n \rangle.$$

Why the condition $f_\alpha s \in \mathcal{O}_X$?

Setup. Let X be a Riemann surface and let D be a Cartier divisor represented by $\{(U_\alpha, f_\alpha)\}$ with $f_\alpha \in \mathcal{M}_X^*(U_\alpha)$ and $f_\alpha/f_\beta \in \mathcal{O}_X^*(U_{\alpha\beta})$.

1) Local-frame (line bundle) viewpoint. On U_α define the local generator (frame)

$$e_\alpha := f_\alpha^{-1}.$$

The intended meaning of $\mathcal{O}_X(D)$ is: "sections are holomorphic multiples of the local frame." Thus a section s on $V \subset U_\alpha$ should be of the form

$$s = h \cdot e_\alpha = h \cdot f_\alpha^{-1} \quad \text{for some } h \in \mathcal{O}_X(V).$$

Rearranging gives

$$f_\alpha s = h \in \mathcal{O}_X(V).$$

Hence the defining condition

$$s \in \mathcal{O}_X(D)(V) \iff f_\alpha s \in \mathcal{O}_X(V)$$

simply says: "s has holomorphic coefficient in the chosen local frame."

2) Order / winding-form viewpoint. For a nonzero meromorphic function g and a point p, let γ_p be a small positively oriented loop. Then

$$\operatorname{ord}_p(g) = \frac{1}{2\pi i} \int_{\gamma_p} \frac{dg}{g}.$$

Using $\dfrac{d(f_\alpha s)}{f_\alpha s} = \dfrac{df_\alpha}{f_\alpha} + \dfrac{ds}{s}$, we get additivity:

$$\operatorname{ord}_p(f_\alpha s) = \operatorname{ord}_p(f_\alpha) + \operatorname{ord}_p(s).$$

Now $f_\alpha s$ is holomorphic near p iff $\operatorname{ord}_p(f_\alpha s) \geq 0$, so

$$f_\alpha s \text{ holomorphic near } p \iff \operatorname{ord}_p(s) \geq -\operatorname{ord}_p(f_\alpha).$$

Thus the condition $f_\alpha s \in \mathcal{O}_X$ is exactly the statement that the pole order of s is bounded by the divisor D.

3) Independence of chart (overlap consistency). On $U_{\alpha\beta}$, write $f_\alpha = u_{\alpha\beta} f_\beta$ with $u_{\alpha\beta} \in \mathcal{O}_X^*(U_{\alpha\beta})$. Then

$$f_\alpha s \in \mathcal{O}_X \iff u_{\alpha\beta}(f_\beta s) \in \mathcal{O}_X \iff f_\beta s \in \mathcal{O}_X,$$

since multiplication by a holomorphic unit preserves holomorphicity (and contributes zero winding/order).

Example on $\mathbf{CP}^1$: $D = n[\infty]$ and $\mathcal{O}(n)$. Let $X = \mathbf{CP}^1$ with charts $U_0 \simeq \mathbf{C}$ (coordinate z) and $U_\infty \simeq \mathbf{C}$ (coordinate $w = 1/z$). For $D = n[\infty]$, take

$$f_0 = 1 \text{ on } U_0, \qquad f_\infty = w^n \text{ on } U_\infty.$$

Then $s \in \mathcal{O}(n)(X)$ iff

$$s \text{ is holomorphic on } U_0 \simeq \mathbf{C}, \qquad w^n s \text{ is holomorphic at } w = 0 \text{ (i.e. at } \infty).$$

Equivalently, $\mathrm{ord}_\infty(s) \geq -n$, so s has a pole at ∞ of order at most n. This forces s to be a polynomial in z of degree $\leq n$, hence

$$H^0(\mathbf{CP}^1, \mathcal{O}(n)) \cong \langle 1, z, \ldots, z^n \rangle.$$

Exercises on $D \sim \deg(D)[\infty]$ on $\mathbf{CP}^1$

Goal

Throughout these exercises, let

$$X = \mathbf{CP}^1 = \mathbf{C} \cup \{\infty\}.$$

The goal is to understand concretely why every divisor D on X is linearly equivalent to

$$\deg(D)[\infty].$$

You should always compute using actual rational functions. Do not skip the order computations at ∞.

Conventions

Use the affine coordinate z on $U_0 = \mathbf{CP}^1 \setminus \{\infty\}$, and the coordinate

$$w = \frac{1}{z}$$

on $U_\infty = \mathbf{CP}^1 \setminus \{0\}$, so that $w = 0$ corresponds to ∞.

For a nonzero meromorphic function f, write

$$(f) = \sum_{p \in X} \operatorname{ord}_p(f)[p]$$

for its principal divisor.

Part I. First computations with divisors on $\mathbf{CP}^1$

1. Compute the principal divisor of z on $\mathbf{CP}^1$. You must compute:

   - $\operatorname{ord}_0(z)$,
   - $\operatorname{ord}_a(z)$ for $a \neq 0, \infty$,
   - $\operatorname{ord}_\infty(z)$ using the coordinate $w = 1/z$.

   Conclude that

   $$(z) = [0] - [\infty].$$

2. For $a \in \mathbf{C}$, compute the principal divisor of $z - a$. Show that

   $$(z - a) = [a] - [\infty].$$

3. Let $a, b \in \mathbf{C}$ with $a \neq b$. Compute the principal divisor of

$$\frac{z-a}{z-b}.$$

Show that

$$\left(\frac{z-a}{z-b}\right) = [a] - [b].$$

Interpret this as saying that any two points of $\mathbf{CP}^1$ are linearly equivalent.

4. Compute the principal divisor of z^n for $n \in \mathbf{Z}$. Be careful about the cases $n > 0$, $n = 0$, and $n < 0$.

5. Compute the principal divisor of

$$f(z) = \prod_{j=1}^{r} (z - a_j)^{m_j},$$

where $a_j \in \mathbf{C}$ are distinct and $m_j \in \mathbf{Z}$. Show that

$$(f) = \sum_{j=1}^{r} m_j [a_j] - \left(\sum_{j=1}^{r} m_j\right)[\infty].$$

Part II. Understanding order at ∞

6. Let $f(z)$ be a polynomial of degree d. Show directly that

$$\mathrm{ord}_\infty(f) = -d.$$

You must rewrite f in the coordinate $w = 1/z$.

7. Let

$$f(z) = \frac{P(z)}{Q(z)}$$

with nonzero polynomials P, Q. Show that

$$\mathrm{ord}_\infty(f) = \deg(Q) - \deg(P).$$

8. Verify explicitly that for every nonzero rational function f on $\mathbf{CP}^1$,

$$\deg(f) = 0.$$

Do this by writing $f = P/Q$, factoring P and Q, and comparing the finite zeros/poles with the order at ∞.

9. Let
$$f(z) = \frac{(z-a_1)\cdots(z-a_r)}{(z-b_1)\cdots(z-b_s)}.$$
Compute (f) explicitly and identify the order at ∞. When does f have a zero at ∞? When does it have a pole at ∞? When is it neither?

10. Compute the divisor of
$$f(z) = \frac{z^2+1}{z(z-1)^3}.$$
List every point where the order is nonzero, including ∞.

Part III. Proving $[a] \sim [\infty]$ and building the general statement

11. Fix $a \in \mathbf{C}$. Using the function $z - a$, prove that
$$[a] \sim [\infty].$$

12. Fix two points $p, q \in \mathbf{CP}^1$. Prove that
$$[p] \sim [q].$$
Do this in two different ways:
- first by reducing to the case $q = \infty$,
- then by explicitly writing a rational function whose divisor is $[p] - [q]$.

13. Let
$$D = \sum_{j=1}^{r} n_j [a_j]$$
with $a_j \in \mathbf{C}$ distinct. Show that
$$D \sim \left(\sum_{j=1}^{r} n_j\right)[\infty].$$
You must produce an explicit rational function f such that
$$(f) = D - \deg(D)[\infty].$$

14. Now let
$$D = \sum_{j=1}^{r} n_j[a_j] + m[\infty].$$
Show that
$$D \sim \deg(D)[\infty].$$
Write out the algebra carefully: do not just say "by degree."

15. Give a complete proof that every divisor D on $\mathbf{CP}^1$ is linearly equivalent to exactly one divisor of the form
$$n[\infty], \qquad n \in \mathbf{Z}.$$
You should prove both:
- existence,
- uniqueness of the integer n.

Part IV. Penetrating exercises: make the equivalence feel inevitable

15. Suppose $D \sim m[\infty]$ and also $D \sim n[\infty]$. Prove $m = n$ without appealing to any abstract theorem about Picard groups. Hint: subtract the two relations and use the fact that principal divisors have degree 0.

16. Show that the map
$$\deg : \mathrm{Div}(\mathbf{CP}^1)/\sim \longrightarrow \mathbf{Z}$$
is a bijection.

17. Let
$$D = [0] + [1] + [2] - 3[\infty].$$
Show directly that $D \sim 0$ by constructing a rational function f with $(f) = D$.

18. Let
$$D = 2[0] - [3] - [\infty].$$
Show that $\deg(D) = 0$, then find an explicit rational function f with $(f) = D$. Do not guess blindly; build f from the required zero/pole data.

19. Let
$$D = 3[1] - 2[i] + 5[-2] - 6[\infty].$$
Is D principal? If yes, construct f explicitly. If no, explain why not.

20. Let
$$D = 2[0] - [1] + 4[\infty].$$
Find the integer n such that $D \sim n[\infty]$, and then write $D - n[\infty]$ as a principal divisor explicitly.

21. Let $p_1, \dots, p_r, q_1, \dots, q_s \in \mathbf{CP}^1$. Give a necessary and sufficient condition for
$$[p_1] + \cdots + [p_r] \sim [q_1] + \cdots + [q_s]$$
on $\mathbf{CP}^1$. Prove your answer.

22. Explain why the statement
$$D \sim \deg(D)[\infty]$$
fails on a general compact Riemann surface X of genus $g > 0$. Your answer should identify exactly what is special about $\mathbf{CP}^1$.

Part V. Exercises through the space $L(D)$

Recall
$$L(D) = \{f \in \mathcal{M}(X)^\times : (f) + D \geq 0\} \cup \{0\}.$$

22. Compute $L(n[\infty])$ for $n \geq 0$. Show that it consists exactly of polynomials of degree at most n:
$$L(n[\infty]) = \operatorname{span}_{\mathbf{C}}\{1, z, z^2, \dots, z^n\}.$$

23. Show that if $D \sim n[\infty]$, then
$$\ell(D) = \ell(n[\infty]) = n + 1 \qquad (n \geq 0).$$

24. Compute $\ell(D)$ for the following divisors:
$$D = [0] + [1], \qquad D = 2[\infty], \qquad D = [0] - [\infty], \qquad D = -[\infty].$$

25. Let $D = [a] + [b] + [\infty]$ with $a, b \in \mathbf{C}$, $a \neq b$. Find a basis for $L(D)$ by first showing $D \sim 3[\infty]$.

26. Let $D = [0] + [1] - [2]$. First compute $\deg(D)$. Then determine a divisor linearly equivalent to D of the form $n[\infty]$. Finally compute $\ell(D)$.

Part VI. Conceptual synthesis

27. Write a short proof, in complete sentences, of the statement:

    On $\mathbf{CP}^1$, two divisors are linearly equivalent if and only if they have the same degree.

    Your proof must use an explicit rational function argument somewhere.

28. Explain geometrically why the point ∞ can "collect" all zeros and poles on $\mathbf{CP}^1$. Your answer should mention the global coordinate z on $\mathbf{C} \subset \mathbf{CP}^1$ and why rational functions have only one extra place where order must be recorded.

29. Suppose a student says:

    "If $D \sim \deg(D)[\infty]$, then all points of $\mathbf{CP}^1$ are literally the same."

    Write a careful reply correcting this statement. Distinguish:

    - equality of points,
    - equality of divisors,
    - linear equivalence of divisors.

30. Compare the statements

$$[a] - [b] = (f) \qquad \text{and} \qquad [a] = [b].$$

    Why is the first usually true on $\mathbf{CP}^1$ while the second is false unless $a = b$?

Challenge problems

31. Let

$$D = \sum_{j=1}^{r} n_j [p_j]$$

    be a divisor of degree 0 on $\mathbf{CP}^1$. Prove that D is principal by constructing a rational function with exactly those zero/pole orders.

32. Give an explicit isomorphism

$$\mathrm{Pic}(\mathbf{CP}^1) \cong \mathbf{Z}$$

    using divisors and degree. Prove that it is well-defined and bijective.

33. Show that

$$\mathcal{O}(D) \cong \mathcal{O}(\deg D)$$

on $\mathbf{CP}^1$. You may use the fact that linearly equivalent divisors define isomorphic line bundles, but you must explicitly connect this to

$$D \sim \deg(D)[\infty].$$

34. Let D be an effective divisor of degree n on $\mathbf{CP}^1$. Show that the complete linear system $|D|$ is the same size as $|n[\infty]|$ after identifying linearly equivalent divisors. Explain this through explicit rational functions.

35. Use the result $D \sim \deg(D)[\infty]$ to give a computation-first proof of the genus-0 case of Riemann–Roch:

$$\ell(D) = \begin{cases} \deg(D) + 1, & \deg(D) \geq 0, \\ 0, & \deg(D) < 0. \end{cases}$$

Instructor notes / hints (optional to include)

- The absolutely central model computation is

$$(z - a) = [a] - [\infty].$$

Everything else should repeatedly reduce to this.

- A good explicit function for a divisor

$$D = \sum n_j [a_j] - \left(\sum n_j\right)[\infty]$$

is

$$f(z) = \prod_j (z - a_j)^{n_j},$$

interpreting negative n_j as poles.

- Students often say "degree zero means principal." Emphasize: this is true on $\mathbf{CP}^1$, but not on higher-genus compact Riemann surfaces.

- To compute order at ∞, always rewrite in the coordinate $w = 1/z$. For example,

$$z - a = \frac{1 - aw}{w},$$

so near $w = 0$ it has a simple pole, hence $\mathrm{ord}_\infty(z - a) = -1$.